

aTOMos v3.6.1.5

UGTS satnav kernel

Passive GNSS positioning
with a source-linked geometric substrate

Corrected-code observations · FP64 QR
UGTS event sequence · Exact 64-bit keys

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Base engine	aTOMos 3.6.1 / M2
Imported prototype	UGTS-KC 3.6.2 / SCLP
Target	RTX 5070 Ti Laptop GPU / 12 GB
CUDA target	sm_120 + compute_120
Release	3.6.1.5 / 14 September 2026

One specialized engine

Observation support and whole-word ASA selection; iterative receiver position and clock-bias estimation; local ENU/log-polar diagnostics; exact source key codecs; UGTS support, compatibility, guard, event, transition and lineage.

The package contains the formal specification, CUDA and CPU source, an independent reference, test inputs and actual verification records. The source's geometric definitions and the new satnav model are distinguished throughout.

Release and implementation record

This release specializes the user's **aTOMos 3.6.1** engine for a concrete satnav workload: **batched passive GNSS code-positioning from prepared observations**. It imports the supplied UGTS SCLP geometry, key and event contracts without claiming that SCLP already contained a pseudo-range receiver model.

Release item	Disposition
Specialization version	3.6.1.5 , as requested; profile SATNAV-R1 .
Imported UGTS source	19 physical PDF pages: three front-matter pages plus 16 numbered main/appendix pages.
Kernel delivered	Native CUDA source with texture and global-read paths; C++17 CPU path.
Navigation computation	Three ECEF position coordinates and one receiver-clock bias per independent epoch.
UGTS integration	Source-style support/compatibility/guard handoff, local chart, time/winding, exact key codecs and lineage records.
Executed here	CPU build/tests, independent Python comparisons, CPU sanitizer checks and local adapter examples.
On-device status	CUDA compile and GPU execution <code>not_run</code> : no nvcc or NVIDIA GPU was available in preparation.

The archive contains a buildable source target, not a prevalidated cubin or Windows executable. Its laptop review script compiles the actual CUDA target, probes the device, compares both input paths against CPU/Python, and records the resulting evidence. A successful CPU run is not reported as a GPU run.

Source keys. U is the attached *UGTS-KC 3.6.2 SCLP* PDF. References to U use its printed page numbers; add three for the physical PDF page. A is the retained 3.6.1 M2 specification/code history in this project. G1–G6 are researched primary GNSS/geodesy references; H1–H4 are NVIDIA hardware/API references. N01–N10 are explicit decisions in this specialization.

The earlier UGTS software archive and the prior aTOMos release archive were not mounted in this run. Their reported test totals are not adopted as tests of the new kernel. The supplied PDF's equations and catalog were read directly. [U, pp. 1–16]

The complete specialized data flow

Prepared GNSS input

Receiver code observations + additive model corrections + satellite positions at emission expressed in reception-time ECEF axes + per-observation uncertainty model.

aTOMos word plane

Ready/present support → ASA → NA → boundary count and whole-word disposition.

Satnav kernel

Selected channels → FP64 geometric ranges and residuals → weighted Givens QR → iterative position/clock solution → final residuals and formal covariance.

UGTS plane

Position candidate → declared local supports → frame/time compatibility → conditional residual interval → computational event → observation record.

Indexed state

ECEF and clock retained in full; ENU, log chart, separate hoop, full time/winding and both 64-bit keys are added to the lineage record. Literal OTAN2 remains a diagnostic.

This interpretation gives the GPU an actual repeated calculation: many independent receiver epochs, replay sessions or test cases. It does not require the display or ray-marching pipeline that U explicitly excludes. [U, pp. 1, 9–12]

What satnav means in this release

The profile computes a receiver-position candidate from already decoded code ranges. A GNSS receiver or upstream software must supply the observations and modelling terms. There is no antenna interface, RF acquisition/tracking loop, signal synthesis, transmitter, route planner or vehicle-control loop in the package. No position is claimed to be measured from a GPU alone.

Position and clock are estimated independently at each epoch. Velocity filtering, carrier-phase ambiguity resolution, RTK/PPP and multi-constellation bias states are separate extensions, not hidden inside the four unknowns of this release.

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1 The imported UGTS and aTOMos contracts

The supplied prototype defines a finite, query-first state-and-relation package. It includes cone/-sphere relations, a bounded translational sweep, log-polar metric and kinematics, explicit time and winding, separate topology profiles, finite grammar, 64-bit key codecs and audited uncertainty/memory claims. Its authoritative handoff is

support → compatibility → guard → verified event → transition → lineage. (1)

This order is retained in the downstream satnav event adapter. [U, pp. 1, 9–12]

1.1 What U does not define

U contains no satellite ephemeris propagation, receiver clock observation equation, code-delay correction model, geodetic datum or GNSS position estimator. These are new SATNAV-R1 definitions, grounded in primary GNSS references rather than attributed to the earlier prototype. [N01]

The base aTOMos contracts retained here are the four-field ASA word rule, explicit JK bit semantics, source-ratio OTAN2, typed unavailable values, immutable input snapshots and ordered result/provenance records. The new matrix in the receiver solve is the explicit *GNSS design matrix*. It is not a silently completed physical beamforming/eigenmatrix from an earlier profile. [A, retained M2 project contract]

1.2 Source figures and key table

U's Figure 4 on printed p. 5 keeps the half-turn state bundle distinct from the orientation-reversing Klein quotient. Its Table 1 and Figure 6 on printed p. 8 keep contiguous packing distinct from MSB-round-robin Morton interleaving. Figure 8 on printed p. 10 returns the result to the canonical event sequence. All three distinctions survive the specialization.

The complete earlier 50-operator catalog remains a source catalog, not a claim that every operator is necessary for positioning. The delivered helpers implement the subset selected below. A source mechanism omitted from the default satnav path is identified rather than repurposed without a declared meaning.

2 Receiver state and source-symbol separation

The authoritative positioning state is

$$z_e = (x_e, y_e, z_e, b_e) \in \mathbb{R}^4, \quad b_e = c \delta t_e, \quad (2)$$

where the first three components are ECEF metres and the fourth is receiver clock bias in metres. Clock seconds are b_e/c with $c = 299792458$ m/s in this profile. This is the standard code-positioning unknown set. [G1]

Quantity	Role	Representation
$\mathbf{r}$	Receiver location	Three unquantized ECEF doubles.
b	Receiver clock bias	Metres, not a branch bit.
P_i	Code observation	Metres, after the explicitly recorded correction convention.
ρ	Local log-radius	UGTS chart quantity, not the code range.
θ	Local angular coordinate	$\text{atan2}(N, E)$ in a fixed ENU chart.
ϕ_h	UGTS hoop phase	Separate supplied periodic field; not silently geographic latitude.
$\varphi_g, \lambda_g, h_g$	Geodetic coordinates	Latitude, longitude and ellipsoidal height.
X, w	Tick and winding	Full integer time identity retained beside its modular key field.
K_C, K_M	Quantized addresses	Two distinct 64-bit codecs, not complete navigation states.

The source's lower-case hoop ϕ is retained as ϕ_h to avoid colliding with geodetic latitude. The earlier aTOMos phase alias is resolved through a named adapter: OTAN2 receives increments of local ρ, θ , while the key retains its separate UGTS hoop field. No 3D coordinate is silently removed. [U, p. 2][N02]

The authoritative records also include epoch identity, time scale, satellite/channel mapping, input/product provenance, statuses and uncertainty assumptions. All fit results preserve the full ECEF estimate and clock value outside the compact key.

3 The corrected-code observation model

A decoded code measurement can be modelled as a geometric range, receiver/satellite clock terms and selected propagation/instrumental delays. SATNAV-R1 accepts an upstream additive correction a_i and defines

$$\tilde{P}_i = P_i + a_i = \|s_i - r\| + b + \varepsilon_i. \quad (3)$$

This is the reduced standard code-positioning model after the declared upstream terms are accounted for. [G1, G4]

For the convention

$$P_i = R_i + b - c\delta t_i^{\text{sat}} + T_i + I_i + B_i + \varepsilon_i, \quad (4)$$

the matching additive correction is

$$a_i = c\delta t_i^{\text{sat}} - T_i - I_i - B_i. \quad (5)$$

The adapter must define the exact code, satellite clock convention, relativistic clock terms and hardware/group biases it uses. The kernel adds the supplied term *once*. It does not repeat an already applied correction or invent a missing ionosphere/troposphere model.

3.1 Satellite coordinates and Earth rotation

s_i is the satellite position at signal emission expressed in the Earth-fixed axes associated with reception. The upstream stage must perform the necessary transmit-time and frame handling consistently. A satellite coordinate computed in emission-time ECEF axes cannot simply be treated as already in the reception axes. [G2]

The solver keeps these supplied positions and corrections fixed during its four-state iteration. The profile is therefore an adapter boundary for prepared code data, not a complete broadcast-navigation decoder. Raw receiver or RINEX values require an upstream modelling adapter before using the named CSV contract. [N01]

3.2 One clock system

There is exactly one receiver-clock unknown. Unaligned measurements from multiple GNSS time systems generally need additional bias treatment; this release does not estimate a hidden inter-system offset. Use compatible code/clock data, or supply and document an upstream alignment under a separate profile.

4 Input epochs, channels and aTOMos selection

Each independent epoch binds its receive-time label, approximate initial state and 32-bit channel masks. Each observation row points to that epoch and names a unique channel in $0, \dots, 31$. Channel IDs, not row order, define mask bits.

Let V contain present channels and X contain channels marked ready. The retained word selection is

$$a = X \wedge M_{\text{ASA}} \wedge V, \quad (6)$$

$$n = a \wedge M_{\text{NA}}, \quad (7)$$

$$h = \text{popcount}(n \wedge M_{\partial}), \quad (8)$$

$$y = \begin{cases} 0, & h > 0, \\ n, & h = 0. \end{cases} \quad (9)$$

The solver uses precisely the channels selected by y . These four fields are emitted unchanged in the result. The default ASA/NA masks are all ones and the default boundary is zero. [A, retained M2 project contract] [N04]

4.1 Whole-word behavior remains literal

A boundary bit intersecting the selected word absorbs the entire epoch group. It does not mean “drop only this satellite.” To explicitly omit an individual observation, use a selector bit, a not-ready flag or an upstream policy with its own provenance. Replacing the boundary behavior with per-bit clearing would change the base engine.

A remaining support of fewer than four channels returns T00_FEW. Four channels must still provide a numerically independent design matrix. A full bit count cannot prove observability by itself. The new GNSS solver checks the resulting matrix rather than treating POPCNT as a complete position solution. [G1][N03]

4.2 Timeline and schema consistency

An epoch identifier binds rows from one receiver event. The time column is full unwrapped GPST seconds under the supplied upstream convention; modular UGTS time is added later. Duplicate epoch IDs, duplicate channels or duplicate satellite IDs within an epoch are errors. Correcting a false upstream time label is not something a CSV parser can infer from equal strings.

The current CSV reader uses finite bounded numeric fields and explicit nonnegative IDs. Physical correction provenance belongs to the input profile; metadata consistency is not a physical synchronization measurement.

5 The weighted positioning step

For current estimate $z^{(k)} = (r^{(k)}, b^{(k)})$, define

$$R_i^{(k)} = \|r^{(k)} - s_i\|, \quad (10)$$

$$v_i^{(k)} = \tilde{P}_i - R_i^{(k)} - b^{(k)}, \quad (11)$$

$$H_i^{(k)} = \left[\frac{x^{(k)} - s_{ix}}{R_i^{(k)}}, \frac{y^{(k)} - s_{iy}}{R_i^{(k)}}, \frac{z^{(k)} - s_{iz}}{R_i^{(k)}}, 1 \right]. \quad (12)$$

Linearization gives $H\Delta z \simeq v$. With declared independent code standard deviations $\sigma_i > 0$, whiten the rows as

$$A_i = H_i/\sigma_i, \quad d_i = v_i/\sigma_i, \quad \Delta\hat{z} = \arg \min_{\Delta} \|A\Delta - d\|_2^2. \quad (13)$$

Then update $z^{(k+1)} = z^{(k)} + \Delta\hat{z}$. The nonlinear model and iterative linearization follow the code-positioning references; the QR implementation is the explicit numerical choice of this specialization. [G1, G4][N03]

5.1 No normal-equation update

The C++/CUDA implementation uses streaming Givens QR on the whitened rows. It does not form and invert $A^T A$ for the update. This avoids squaring the design matrix's condition number as part of the solution method. The independent Python reference uses a different Householder QR implementation on the full row matrix.

All four unknowns use metres, including the clock bias. A floating-point conversion to seconds is applied only to a displayed/consumer clock offset. This keeps a factor of the speed of light out of the fourth Jacobian column.

The default iteration stops when both position-increment norm and absolute clock increment are at most 10^{-4} m, with at most 12 steps. These are numerical convergence controls, not a statement that real measurements yield 0.1 mm accuracy. The convergence domain depends on the input model and initial approximation.

6 Streaming QR, rank and formal covariance

The kernel maintains an upper-triangular 4×4 matrix R and a transformed right-hand side z . Each whitened row is appended by rotations that eliminate its components against the current triangular diagonal.

For diagonal R_{jj} and current row component a_j , let

$$\gamma = \text{hypot}(R_{jj}, a_j), \quad c_j = R_{jj}/\gamma, \quad s_j = a_j/\gamma. \quad (14)$$

The same rotation is applied to the remaining row and right-hand-side entries. A zero pair needs no rotation. The result is solved by back substitution.

6.1 The implemented rank screen

The retained check is

$$|R_{jj}| > \tau \max_k |R_{kk}| \quad \text{for all } j, \quad \tau = 10^{-10}. \quad (15)$$

This is a declared unpivoted triangular-diagonal screen. It detects the tested rank-deficient cases, but is not an SVD condition-number certificate or a proof of well-conditioned geometry for all possible inputs. A failed screen produces RANK_DEFICIENT, not a coordinate filled with zeros. [N03]

6.2 Covariance is tied to the assumed sigmas

After convergence the Jacobian is rebuilt at the final estimate. From R^{-1} , compute the diagonal of

$$P = (H^T W H)^{-1} = R^{-1} R^{-T}, \quad W = \text{diag}(\sigma_i^{-2}). \quad (16)$$

The implementation obtains this from triangular solves, not a dense matrix inverse of $H^T W H$. It returns coordinate/clock variances in squared metres. This is formal uncertainty under the declared local linear and noise model. [G5]

It is not measured error against truth, and not a guaranteed bound on faulty or mis-modelled measurements. The independent fixture comparisons check both estimates and these formal diagonal values; true-error metrics are separately computed only because the artificial fixture has a known generating position.

7 Solver status and residual diagnostics

Convergence, residual fit and the later UGTS event are separate fields.

Solver status	Meaning
CONVERGED	Numerical increment thresholds reached with valid final calculations.
MASK_ABSORBED	Selected word intersected the retained whole-word boundary.
TOO_FEW	Fewer than four selected observations.
RANK_DEFICIENT	Triangular rank/covariance check failed.
ITERATION_LIMIT	The selected number of iterations did not satisfy convergence.
NUMERIC_FAILURE	Invalid range, nonfinite result or unusable numeric intermediate.

For a converged solution, preserve every selected post-fit residual $r_i = \tilde{P}_i - \|s_i - \hat{r}\| - \hat{b}$, and report

$$\text{RMS} = \sqrt{\frac{1}{m} \sum_i r_i^2}, \quad S = \sum_i (r_i/\sigma_i)^2, \quad Z_{\max} = \max_i |r_i/\sigma_i|. \quad (17)$$

The default comparison $Z_{\max} \leq 6$ is a named residual budget. It is not a leverage-normalized test or a claimed false-alarm probability. The solver does not automatically discard a channel and retry until this flag passes. [N09]

Exactly four observations give NO_REDUNDANCY, even if their residuals are small. For more observations, the flag is WITHIN_RESIDUAL_BUDGET or RESIDUAL_ALERT. A single gross-outlier fixture converges numerically but produces the latter alert; the data record retains its large actual position error.

RAIM/ARAIM, HPL/VPL, continuity risk, certified spoof detection and navigation-action authority are not supplied by these flags. A valid solution record can be useful for replay and analysis without representing those additional guarantees.

8 Local ENU and the UGTS log chart

The receiver solution remains three-dimensional. For a declared geodetic anchor $(\varphi_0, \lambda_0, h_0)$, convert ECEF differences to East/North/Up using

$$R_{ENU} = \begin{bmatrix} -\sin \lambda_0 & \cos \lambda_0 & 0 \\ -\sin \varphi_0 \cos \lambda_0 & -\sin \varphi_0 \sin \lambda_0 & \cos \varphi_0 \\ \cos \varphi_0 \cos \lambda_0 & \cos \varphi_0 \sin \lambda_0 & \sin \varphi_0 \end{bmatrix}. \quad (18)$$

Then $(E, N, U)^T = R_{ENU}(\mathbf{r} - \mathbf{r}_0)$. The transform is an explicit GNSS/geodesy addition, separate from the source's abstract local chart. [G3]

With horizontal radius $r_h = \sqrt{E^2 + N^2}$, use

$$\rho = \ln(r_h/r_0), \quad \theta = \text{atan2}(N, E). \quad (19)$$

Up remains a separate value. A declared core radius returns `origin_core` instead of evaluating a singular logarithm. Out-of-range ρ is not automatically wrapped into a different geographic position. [U, pp. 3–4][N02]

8.1 Metric and finite steps carried from U

For the planar chart,

$$ds^2 = r_h^2(d\rho^2 + d\theta^2), \quad \Delta r_h = r_h(e^{\Delta\rho} - 1), \quad (20)$$

$$J = r_h \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (21)$$

These identities explain the local representation; the new satellite-range solve still uses full ECEF coordinates. Quantizing this chart does not improve the original measurement precision or create a third coordinate. [U, p. 4]

The geodetic utility uses the declared WGS84 ellipsoid constants $a = 6378137$ m and $1/f = 298.257223563$. Its height is ellipsoidal height, not an unprovided geoid-corrected height above mean sea level. The reference tests its conversions and inverse relation over a finite set of points. [G6]

9 OTAN2 in satnav: preserve the observable

The inherited literal operator remains

$$\text{otan2}_s(\Delta\theta, \Delta\rho) = \arctan(\Delta\theta/\Delta\rho), \quad \Delta\rho \neq 0, \quad (22)$$

with a separately supplied reference-axis subtraction when selected. This is a *log-chart motion slope*. It is not used to solve pseudoranges, and is not relabeled as the receiver's geographic heading or satellite azimuth. [A, retained M2 project contract]

The passive adapter compares successive defined local chart records in the same frame. It uses an explicitly named principal-difference policy for $\Delta\theta$; that policy does not reconstruct missed revolutions. Full linear time and the chosen phase convention are stored with the observation.

9.1 Missing direction remains a status

A first observation has no previous chart increment. A zero pair has no direction. A pure angular increment has an undefined literal ratio. The adapter records `first_observation`, `zero_increment` or `ratio_undefined` rather than inventing a perfect zero angle. A sequence with non-increasing time is identified instead of being silently reordered into a new physical history.

A change in the axis alone changes the raw residual by the opposite angle. A common positive scaling of both increments preserves the ratio, and simultaneous reversal also preserves the literal ratio while reversing the actual directed motion. These are inherited operator laws, not new navigation-sensor guarantees.

9.2 No new J/K feedback hidden in the adapter

The aTOMos J–K equation is retained:

$$q^+ = (J \wedge \neg q) \vee (\neg K \wedge q). \quad (23)$$

The supplied adapter uses the explicit hold event $J = K = 0$ for its local example state. It does not turn a coordinate, residual or support bit into a steering action. The encoded circle-plus blend and the independent alignment/limit/fringe predicate roles remain part of the base contract; no unprovided bit encoder is guessed here.

This is a specialized measurement-and-query kernel. The source's broader kinematic and control vocabulary does not introduce an unrequested external action interface.

10 Paired supports and the canonical UGTS event

U normalizes the two spherical supports as

$$f_L(x) = \|x - c_L\| - R_L, \quad f_R(x) = \|x - c_R\| - R_R, \quad (24)$$

with a profile selecting union, intersection or either-with-tag. The overlap flag is retained separately. [U, p. 3]

SATNAV-R1 uses the union of two *declared local processing domains* for its worked event adapter. With supplied deterministic position-radius allowance u_p , its per-sphere margin is

$$m_j = R_j - \|x - c_j\| - u_p. \quad (25)$$

Admission means at least one margin is nonnegative. This is a geometric/query policy for the supplied local frame, not a claim that the true receiver error is bounded by u_p . Its origin is explicitly recorded. [N09]

The corresponding handoff is implemented in order:

UGTS stage	Satnav realization
Support	Candidate lies within at least one declared local support under the supplied allowance.
Compatibility	Fixed frame and increasing unwrapped time; supported estimator-clock convention.
Guard	Residual interval compared with a named metre-domain budget.
Verified event	Candidate satisfies those computational/model conditions with residual redundancy.
Transition	Publish an observation record only. No external control command.
Lineage	Source/profile IDs, full time, source keys and the preceding record digest.

For the demo, $u_p = 5$ m, the two support radii are 10,000 m, and centres are (0, 0, 0) and (100, 100, 0) ENU metres. These are labelled worked parameters, not values extracted from the attached prototype or a real site survey.

The phrase POSITION_CANDIDATE_VERIFIED means verified under this declared computational handoff. It does not supply certified GNSS integrity or prove the physical correctness of the upstream observations.

11 Finite cones and translational sweeps retained

For the source's slant length $T > 0$ and half-angle $0 < \alpha_c < \pi/2$,

$$h = T \cos \alpha_c, \quad R = T \sin \alpha_c. \quad (26)$$

With apex c , unit axis a and query x , set

$$z = a \cdot (x - c), \quad q = \|(x - c) - za\|. \quad (27)$$

The exact primitive relation is the signed meridian distance to the triangle $(-R, h), (R, h), (0, 0)$, with negative interior and zero boundary. The implementation computes distances to the three boundary segments and uses the source's inside test $0 \leq z \leq h, q \leq z \tan \alpha_c$. [U, pp. 2–3]

These are optional UGTS query helpers in the archive. A finite cone is not silently substituted for the satellite pseudorange sphere or a physical antenna response. The default receiver solve uses the actual distance-to-satellite equations already stated.

For a fixed-orientation cone translated along a segment of length L , $n \geq 2$ uniform samples and sample minimum m_n , retain

$$m_n - \frac{L}{2(n-1)} \leq \inf_{u \in [0,1]} d_C(x - s(u)) \leq m_n. \quad (28)$$

The source derives this using the one-Lipschitz distance response to translation and the nearest sample spacing. It is not a general exact SDF for arbitrary rotated sweeps. [U, p. 3]

11.1 Numerical and nesting distinction

The helper evaluates this analytic interval in ordinary floating arithmetic. It does not implement directed rounding or a proven machine-error enclosure. The interval's *width* decreases with sample spacing. Intervals from arbitrary nonnested sample grids need not be nested, so the implementation does not claim that stronger property. [N07]

The source's constraint-release and hinge-torque mechanisms remain separately typed geometric/mechanical profiles. Removing a constraint is not an automatic improvement to satellite observability, nor does the existence of a hinge model supply a physical receiver trajectory. They are not inserted into the default position estimator.

12 Jitter intervals and the explicit source correction

U defines $f_j = f + \epsilon_j \sigma$ with $\sigma \in \{-1, +1\}$, then prints the equality

$$[f_j - \epsilon_j, f_j + \epsilon_j] = [f - \epsilon_j, f + \epsilon_j]. \quad (29)$$

That equality is generally false for nonzero ϵ_j . This edition preserves its source location and explicitly separates the two interpretations rather than silently changing the supplied page. [U, p. 5][N06]

12.1 Two valid, different interval statements

The perturbation family obeys

$$f_j \in [f - \epsilon_j, f + \epsilon_j]. \quad (30)$$

If f_j is the available perturbed observation and $|f_j - f| \leq \epsilon_j$, then

$$f \in [f_j - \epsilon_j, f_j + \epsilon_j]. \quad (31)$$

Both are valid enclosure statements, but the interval centres differ. The helper returns both named intervals and a test verifies that they are not falsely equated. The source's example centred around f is treated as the perturbation-family interval.

No jitter is added to the GNSS code observations or to solved ECEF coordinates. The deterministic bit can be metadata or a separate query perturbation, not a method for manufacturing measurement precision or removing multipath.

12.2 Metre-domain guard in the event adapter

For the largest absolute post-fit residual $m = \max_i |r_i|$ and a *supplied* absolute residual perturbation allowance u_r , form

$$[\max(0, m - u_r), m + u_r]. \quad (32)$$

A budget B is met when the upper endpoint is at most B , exceeded when the lower endpoint is greater than B , and otherwise unresolved. This is a conditional model classification. The demo uses $u_r = 0.1$ m and $B = 20$ m as stated fixture parameters. It does not identify nominal σ_i with a hard worst-case bound.

13 Two exact 64-bit key layouts

U's field widths and layouts are retained exactly:

Field	Bits	Contiguous range	Domain
Log-radius index	20	63:44	Declared $\rho \in [-20, 0]$
Chart angle	18	43:26	Periodic $[0, 2\pi)$
Modular tick	14	25:12	$0, \dots, 16383$
Hoop phase	12	11:0	Periodic $[0, 2\pi)$

The contiguous word is

$$K_C = (q_\rho \ll 44) \vee (q_\theta \ll 26) \vee (q_X \ll 12) \vee q_{\phi_h}. \quad (33)$$

The Morton word emits source bits in MSB round-robin order

$$\rho_{19}, \theta_{17}, X_{13}, \phi_{11}, \rho_{18}, \theta_{16}, X_{12}, \phi_{10}, \dots \quad (34)$$

until each source field is exhausted. They are intentionally different encodings. [U, pp. 7–8]

13.1 The source's exact golden pair

For (949111, 0, 1920, 227), the new tests recover

```
contiguous: e7b77000007800e3
Morton:    88823bb88099128b
```

Both decode to the same quantized tuple. Random round trips and prefix-refinement bounds are tested separately from the new positioning arithmetic.

13.2 Address identity is not state identity

The 64-bit key has exactly 2^{64} possible words. A key does not encode arbitrary clock bias, full linear time, Up, formal covariance, correction products or lineage. Those remain separate full fields in the event record. The 12-bit field is the named hoop phase, not a simultaneously hidden arbitrary matrix and grammar payload. [U, pp. 8–9][N05]

The address construction follows the position estimate; it is not used to quantize satellite coordinates or code ranges before the receiver solve. A matching key is an index match, not proof that two entire navigation events are identical.

14 Quantization, error and explicit origin handling

SATNAV-R1 declares nearest-node radial quantization over $[-20, 0]$:

$$h_\rho = \frac{20}{2^{20} - 1}, \quad q_\rho = \text{round}_+ \left(\frac{\rho + 20}{20} (2^{20} - 1) \right). \quad (35)$$

Angles use nearest periodic nodes with upward half ties, while the source's bit widths and range remain unchanged. This tie/quantization choice is an explicit specialization, not an unquoted algorithm recovered from an unavailable source ZIP. [N05]

For an angular width b , $h_\theta = 2\pi/2^b$. A nearest-node quantization gives $|\delta\rho| \leq h_\rho/2$ and circular $|\delta\theta| \leq h_\theta/2$ within the chosen range. Using the exact source radial step, a conservative planar error bound is

$$\|\hat{x} - x\| \leq r_h(e^{h_\rho/2} - 1) + 2r_h e^{h_\rho/2} \sin(h_\theta/4). \quad (36)$$

It follows from adding a radial displacement and an angular chord at the enlarged radius. The bound scales with radius; log packing is not uniform physical resolution. [U, pp. 4, 8–9]

14.1 Default diagnostic chart

The demo chooses $r_0 = 100000$ m and a 0.1 m origin core. It keys only values with $-20 \leq \rho \leq 0$. These choices define an indexed diagnostic domain, not the receiver's positioning accuracy or a geographic route. Out-of-range or core values keep the full position and a chart status without wrapping them into a new position.

14.2 No equal-information compression claim

U explicitly audits the ratios 192/64, 32/1 and 512/12 as nominal width ratios between records with different semantics. This specialization does not reintroduce a claim that one address is equivalent to all original floating information. [U, pp. 8–11]

The packed key is useful for lookup and local indexing after the relevant state exists. Position estimation, corrected observations, uncertainty and event verification remain real computational and storage costs.

15 Linear time, winding and separate topology profiles

For reference tick X_0 , period $P > 0$ and full linear tick X ,

$$u = \frac{X - X_0}{P}, \quad w = \lfloor u \rfloor, \quad \psi = u - w \in [0, 1). \quad (37)$$

The source's example $X = 1920$, $X_0 = 1135$, $P = 256$ yields $w = 3$ and $\psi = 0.06640625$. The new helper reproduces that value, including floor-division behavior for negative ticks. [U, p. 5]

The satnav key stores only the 14-bit tick residue, while the event retains full tick, GPST coordinate, origin/period and winding. A period of 16384 one-second ticks repeats its residue after 4 h 33 min 4 s. It does not identify events across those repetitions without the additional time/lineage fields.

15.1 The source half-turn is not the reflective quotient

U's literal odd radial-wrap bundle applies

$$(\rho, \theta, \phi_h, o) \mapsto (\rho', \theta + \pi, -\phi_h, -o). \quad (38)$$

Its separately named reflective Klein profile applies

$$(\rho, \theta, \phi_h, o) \mapsto (\rho', \pi - \theta, -\phi_h, -o). \quad (39)$$

The second reverses angular orientation; the first is a rotation on the base circle with a changed internal state. The archive retains independent helpers and tests, matching the distinction in U's Figure 4. [U, pp. 5–6]

Neither rule is applied to the GNSS receiver's ECEF position, a satellite orbit or the physical Earth. A local index experiment may select either profile explicitly; it does not make a geographical discontinuity an actual displacement. [N08]

The source's finite grammar and radix refinement remain local state/index mechanisms. Their existence does not turn one receiver solution into an unbounded tree or prove that every possible trajectory is retrieved in constant time.

16 GPU kernel: one independent solve per thread

The kernel takes a bounded batch of immutable epoch metadata and a structure-of-arrays observation buffer. One thread evaluates one full independent four-state solution. Each thread owns its QR workspace, iterative estimate, status and residual record. There is no cross-epoch update or unordered reduction of position states.

```
const size_t e = size_t(blockIdx.x) * blockDim.x + threadIdx.x;
if (e < count) {
    output[e] = solve_epoch(epochs[e], observations, e, config);
}
```

The launch uses 128 threads per block. Repeated jobs can supply thousands of independent epochs; a single live low-rate receiver does not by itself provide enough parallel work to establish a GPU speed advantage. This kernel is a batch/replay specialization.

16.1 Numerical implementation

The actual shared implementation is in `include/satnav_core.hpp`. It performs the retained ASA arithmetic, row construction, QR update, convergence and final covariance/residual calculation. The CUDA translation unit invokes that implementation with its device reader. The CPU uses the same word and numerical function with a plain double reader.

An independent Python implementation uses separate matrix construction and Householder QR. It is not merely calling back into the shared C++ function. The two verification levels therefore distinguish device/host conformance from an additional numerical reference comparison.

16.2 Potential performance costs remain visible

Independent threads may finish in different iterations, producing normal warp imbalance. QR uses FP64 arithmetic and square roots; no tensor-core speed claim is made. The source keeps full precision where satellite-range subtraction requires it, rather than forcing coordinates into float textures or one-bit words.

The build disables fused contraction for the shared review profile and does not enable fast math. The acceptance checks use named tolerances, not a claim of bit-identical floating evaluation across every compiler and processor.

17 Texture and global input paths

For N independent epochs, channel $c \in [0, 31]$ and observation field $f \in [0, 5]$, flatten the input as

$$\text{index}(e, c, f) = (6c + f)N + e. \quad (40)$$

At a fixed channel/field, neighboring epoch threads therefore read neighboring values. The six fields are satellite x, y, z , raw code, additive correction and nominal sigma.

17.1 Exact FP64 bit transport in the texture path

A CUDA linear texture resource uses an int2 channel. Each texel carries the 64 raw bits of one double; the device reconstructs the value using the low and high 32-bit words. The read uses an integer `tex1Dfetch`; there is no interpolation, normalized-coordinate lookup or conversion of ranges into float values. [H4][N10]

```
const int2 bits = tex1Dfetch<int2>(texture, integer_index);
const double value = __hiloint2double(bits.y, bits.x);
```

The global path reads the same double buffer directly. Both paths use the same solver and output record. Equality of decoded inputs is a data-representation property; it does not guarantee a texture speedup for this access pattern.

17.2 Ownership and publication

Device buffers and events use local ownership wrappers. A texture object is destroyed before its backing buffer. Each launched epoch has one writer. The program checks CUDA allocation, copies, launch and completion before publishing downloaded result files. A failed device call is an error, not a CPU fallback labelled as a GPU success.

The first chunk receives an untimed warmup. Device events measure subsequent kernel intervals and their sum is reported. CSV parsing, allocations, texture construction, host packing, uploads, downloads and the Python event adapter are outside that sum. A CPU loop interval and a full application latency are therefore distinct quantities.

18 The RTX 5070 Ti Laptop target and VRAM plan

NVIDIA lists the **RTX 5070 Ti Laptop GPU** with a **12 GB GDDR7** memory configuration. NVIDIA's CUDA table lists the GeForce RTX 5070 Ti family at compute capability 12.0; the build selects `sm_120` and matching virtual code. CUDA 12.8 introduced compiler support for `SM_120`. These are supporting implementation facts, not equations supplied by the UGTS prototype. [H1–H3]

The program's `-probe` path records the actual device name, major/minor capability, driver/runtime versions, total/free memory and linear-texture limit. This edition's targeted binary reports a mismatching capability rather than assuming that a marketing label guarantees the compiled architecture.

18.1 Payload accounting

The executed CPU build reports 72 bytes per epoch metadata record and 376 bytes per result. The fixed observation representation is $32 \cdot 6 \cdot 8 = 1536$ bytes per epoch, so the tested ABI payload is

$$M_{\text{payload}}(N) = N(72 + 376 + 1536) = 1984N \text{ bytes.} \quad (41)$$

The 256-epoch demo therefore uses 507,904 payload bytes. A maximum chunk of 65,536 epochs uses 124 MiB before a separate 64 MiB planning margin. These figures do not count the entire driver/context or all host copies.

Admission uses the actual free/total bytes, the user budget and reserve. The defaults are a 512 MiB budget and 1536 MiB total-device reserve. After checking that each subtraction is valid, capacity is bounded by

$$B_{\text{usable}} = \min(B_{\text{user}}, 0.7F, T - R_{\text{reserve}}) - 64 \text{ MiB.} \quad (42)$$

The integer implementation rounds the free-memory fraction downward, caps each chunk at 65,536 epochs and checks texture/index limits. Another process may consume memory after probing, so actual allocation errors still propagate.

The package does not fill all 12 GB to demonstrate activity. Useful payload depends on the batch; automatically leaving space for other laptop processes is a resource plan, not a claim of permanent texture-cache residency. No operating-system ring-0 driver or executable-code-in-texture mechanism is part of the CUDA interface.

19 Actual input and result files

The executable reads two CSV files with exact headers. The source-to-domain model is recorded separately in `profile.json`; prepared real data must supply its own correction/product provenance.

```
epochs.csv:
epoch_id,t_rx_gpst_s,x0_m,y0_m,z0_m,b0_m,
asa_mask,na_mask,boundary_mask

observations.csv:
epoch_id,channel,satellite_id,sx_rx_m,sy_rx_m,sz_rx_m,
code_m,add_correction_m,sigma_m,ready
```

The display above wraps the headers for readability; the actual files use one header line each. `docs/INPUT.md` defines their exact names, ranges and physical meaning. The reader rejects malformed headers, duplicate channels, nonfinite fields and invalid sigmas before passing data to a backend.

19.1 Results

`solutions.csv` retains one row per input epoch: source/time identity, solver and fit status, iteration count, channel count, all four ASA fields, position/clock, formal diagonal covariance and residual statistics. `residuals.csv` records each active post-fit channel when the solver converges.

For a failed solve, estimate fields may contain the starting guess or a partial iterate. The status is authoritative; those fields must not be published as a valid fix. Zero variances in an unevaluated result do not mean a perfectly known position. The parser and status record prevent that interpretation without hiding the original calculation state.

`run.json` records the backend, timing scope, selected numerical settings and actual device report where available. `ugts_events.jsonl` and `ugts_summary.json` are produced by the explicit post-processing step. They do not replace the original solution or residual files.

A new output directory is required for each run. Completed runs can be compared without silently overwriting earlier evidence. Corruption of a source file or a changed result is visible in the delivered integrity manifest or retained event chain.

20 Ordered event records and source lineage

A UGTS event binds the solved candidate, coordinate transforms, full time, optional keys, support-/guard record and previous digest. The source's ordering is retained, while the satnav-specific meanings are labelled explicitly.

```
{
  "version": "3.6.1.5",
  "quantity": "receiver_position_candidate",
  "epoch_id": 0,
  "solver_status": "CONVERGED",
  "ecef_m": ["three retained numeric components"],
  "clock_bias_m": "retained numeric clock bias",
  "enu_m": ["E", "N", "Up retained separately"],
  "chart": {"rho": "...", "theta": "...", "hoop_phi": "..."},
  "packing": {"contiguous_hex": "...", "morton_hex": "...",
    "linear_tick": 0, "winding": 0},
  "support": {}, "compatibility": {}, "guard": {},
  "event": "POSITION_CANDIDATE_VERIFIED",
  "transition": "publish_observation_only",
  "previous_hash": "...", "record_hash": "...
}
```

This is a structural sketch; the complete worked exports contain actual numeric values and statuses, not these explanatory strings.

The chain uses a domain-tagged SHA-256 of canonical local JSON and retains the prior head. It binds the emitted sequence to a retained reference. It is not a signature, a satellite-signal authentication primitive or proof that an upstream observation was physically true.

Author attribution is included as supplied: Tom Klootwijk, with the identifier and date in the release metadata. A full profile/source digest remains separate from the 64-bit coordinate key or a one-bit event field. This preserves the base engine's identity/state distinction and U's requirement that uncertainty and lineage survive packing. [A, retained M2 project contract] [U, pp. 8–10]

21 The deterministic worked satnav batch

The shipped demo has 256 artificial receiver epochs. The anchor is a declared WGS84 location at 52° latitude, 5° longitude and 100 m ellipsoidal height. Receiver points are generated from a local ENU path, with a known clock bias near 75,000 m. Satellite directions and ranges are constructed fixtures, not a downloaded constellation or an observation session from the user's laptop.

The first 128 cases are noiseless model data. The next ordinary cases add independent Gaussian fixture noise with their specified sigmas. The final six cases deliberately exercise failure and edge conditions. Corrections are labelled artificial additive test values, not an atmosphere or ephemeris model.

Outcome	Observed count / result
Converged numerical epochs	253 of 256
Too few selected satellites	1
Whole-word absorbed	1
Rank-deficient geometry	1
Converged within chosen residual budget	251
Gross-outlier residual alert	1
Exactly four, no residual redundancy	1

The exact-four and 32-channel cases verify those interface boundaries. The outlier case adds an artificial 200 m error to one code observation. It converges to an incorrect candidate with a large residual alert, demonstrating why convergence and navigation reliability are not the same field.

The geometry-support adapter then emits 251 computationally verified candidate records, three no-solution records and two unresolved candidates under its declared conditions. The exact-four case remains unresolved because no redundancy is present; the outlier remains unresolved because the chosen residual criteria are exceeded.

These are finite results of a known model. The demo is useful for reviewing signs, units, frame adapters and implementation agreement. It is not a real-world accuracy measurement or a GPU speed result.

22 Independent results and error measures

The C++ solve is compared against a separately written Python Householder solver. The comparison checks epoch identity, all mask fields, status/fit, position and clock, formal covariance and residual statistics, yielding **4,828 checked comparisons** over the 256 epochs.

Cross-reference quantity	Observed maximum
Absolute state-component difference	$1.210719347 \times 10^{-8} \text{ m}$
Formal covariance diagonal difference	$6.394884622 \times 10^{-14} \text{ m}^2$

The comparison tolerance for each state component is 10^{-4} m . Small algorithmic agreement does not make the underlying physical inputs exact; it shows that the two implementations solve the same declared finite problem to closely matching values.

Fixture class	Epochs	3D position RMSE	Maximum error
Noiseless	128	$3.61 \times 10^{-9} \text{ m}$	$7.84 \times 10^{-9} \text{ m}$
Noisy	122	1.6017 m	3.3883 m
Gross outlier	1	147.0987 m	147.0987 m
Exactly four	1	$1.23 \times 10^{-8} \text{ m}$	$1.23 \times 10^{-8} \text{ m}$
32 channels	1	$5.71 \times 10^{-9} \text{ m}$	$5.71 \times 10^{-9} \text{ m}$

Errors above are measured only against the synthetic generating truth. The nanometre- scale noiseless figures reflect arithmetic roundoff in matched artificial data, not an achievable GNSS hardware accuracy. The noisy fixture demonstrates the selected noise model; a real receiver's error includes its actual correction/model limitations.

Formal covariance, predicted geometry quality and empirical truth error are different quantities. The package keeps them separate, following the distinction made in the positioning-error reference. [G5]

23 Verification completed and pending

Category	Recorded result
CPU C++ build and CTest	Passed
C++ explicit checks	105,559 passed
Python UGTS/geodesy tests	18 passed
Command-line scenarios	11 passed
Independent solver comparisons	4,828 passed
CPU Address/UndefinedBehavior Sanitizer	CTest passed
CUDA compilation	not_run: nvcc unavailable
CUDA execution / memory sanitizer	not_run: no accessible NVIDIA GPU
Windows native build	not_run in preparation
Live GNSS observation validation	Not performed; synthetic fixtures only

The C++ count includes all 65,536 possible 16-bit population-count cases and 10,000 independent random word-mask comparisons. It also checks the full JK truth table, QR identities, covariance, rank rejection and the retained whole-word rule. A large assertion count is not a count of separate receivers tested.

The Python suite checks both key codecs against the source's golden pair and 5,000 random tuples, all prefix lengths for the reference key, quantization bounds, full winding, two separate topologies, cone/sweep relations, the jitter-interval distinction, geodetic/ENU round trips and literal OTAN2 semantics.

The CLI scenarios include malformed IDs/channels, negative sigma, NaN input, invalid back-end/options, existing-output protection, a normal run and an iteration-limit run. All outcomes are recorded under `results/`, with the earlier source test claims kept separate from these actual executions.

The device validation script has already been invoked in this environment and records its unavailable-tool result. On the laptop it becomes the required next validation step, not a placeholder success. No compiled CUDA speedup or texture-residency claim is made in this edition.

24 Build, run and verify on the requested laptop

Use a CUDA-supported host compiler, CMake 3.24 or newer, and CUDA Toolkit 12.8 or newer for the requested SM120 target. The installed NVIDIA driver must support the chosen runtime/-toolchain. This specifies a minimum architecture-capable toolkit, not a claim that 12.8 is the newest available release. [H2, H3]

```
cmake -S . -B build_gpu -DSATNAV_ENABLE_CUDA=ON \
  -DCMAKE_BUILD_TYPE=Release
cmake --build build_gpu --config Release --parallel 2
ctest --test-dir build_gpu -C Release --output-on-failure
./build_gpu/satnav --probe
./build_gpu/satnav --input examples/demo --out run_gpu \
  --backend cuda --read texture --verify
python tools/verify_run.py --input examples/demo --run run_gpu
python tools/ugts_handoff.py --input examples/demo --run run_gpu
```

On a Windows multi-configuration build, use the following executable path:

```
build_gpu/Release/satnav.exe
```

Run configuration from the matching x64 compiler developer terminal so that CMake can locate the host toolchain.

24.1 Complete device comparison

```
python tools/validate_gpu.py --sanitizer
```

The script records compilation, device probing, texture/global runs, shared CPU comparisons, independent Python comparisons and requested memcheck results. It uses fresh run directories. Missing required tooling produces a separate unavailable status and exit code 3. A real build/test error remains a failed status.

24.2 CPU-only review

```
cmake -S . -B build_cpu -DCMAKE_BUILD_TYPE=Release
cmake --build build_cpu --config Release --parallel 2
./build_cpu/satnav --input examples/demo --out run_cpu
python tools/verify_run.py --input examples/demo --run run_cpu
python -m unittest discover -s tests -p "test_*.py" -v
```

The Python reference and adapters use the standard library. No PyTorch or external model service is required for this kernel.

25 Integration decisions and source issue register

ID	Decision	Effect
N01	Corrected-code satnav model	The GNSS observation equation and solver are explicit new additions, not attributed to U.
N02	Full receiver state retained	Solve in 3D ECEF plus clock; use a separate local log chart and retain Up and provenance.
N03	Independent QR methods	Shared CPU/CUDA Givens implementation and independent Python Householder solve; named rank and convergence policy.
N04	Fixed channel-to-bit mapping	One epoch is one at-most-32-channel word. Boundary absorption stays whole-word.
N05	Source key layouts preserved	Exact contiguous/Morton pair retained; quantization ties named; key does not replace full state.
N06	Two jitter intervals separated	Explicit source equality issue documented; perturbation-family and measurement-centred intervals are not silently equated.
N07	Bounded sweep interpretation	Translation-only analytic interval; floating arithmetic and nonnested grids not promoted to stronger certificates.
N08	Topology does not rewrite Earth	Bundle/reflection helpers remain separate local contracts, not GNSS propagation rules.
N09	Computational event semantics	Declared support/residual policies produce observation records, not a protection level, spoof-proof fix or actuator command.
N10	Honest hardware evidence	CUDA source targets the requested laptop; on-device build/test status stays unavailable until actually run.

These decisions define one specialized profile of aTOMos 3.6.1 with the supplied UGTS prototype integrated at explicit interfaces. They do not discard the user's operator vocabulary or pretend that all missing GNSS physics were already encoded in a name.

A later variant can extend the estimator with new states, observable types or correction adapters. It must identify the changed observation model and preserve a projection or compatibility test to this profile where appropriate. An unreported inter-system bias, new residual exclusion rule or changed topology is not a transparent optimization.

26 Source and artifact inventory

U. *UGTS_KC_3_6_2_SCLP_Tom_Klootwijk(1).pdf*. The file has 19 physical pages and a 16-page numbered main/appendix sequence. The source's referenced 13-page precursor and software ZIP were not supplied here. Its current PDF SHA-256 is

```
6987490a0a7b7ed76b0091057a2e69ab831fee786434a0b4219782923f5ea580
```

The U page references in this formalization point to the printed main numbers. For example, its key table on printed p. 8 is physical PDF page 11. The relevant figure, table and jitter-equation pages were inspected as rendered pages as well as text.

A. aTOMos v3.6.1 / M2 source and code contract in this project conversation. The prior archive was not mounted in this turn; the specialization preserves the written ASA/JK/OTAN2 contracts rather than claiming a binary patch to an unavailable artifact. The actual new source is included and tested to the extent recorded above.

Delivered file family	Purpose
include/, src/, cuda/ python/	Shared solver, executable and real CUDA backend source. Independent solution and source-derived UGTS/geodesy helpers.
examples/demo/ tests/, results/ tools/	Prepared artificial epochs, observations, truth and profile. Reproducible checks and the actual recorded outcomes. Build/review, independent comparison, event adapter and integrity tools.
docs/	This editable formalization, input contract and source-decision register.
source/manifest.json	Exact source availability and fingerprint record.
SHA256SUMS.txt	Delivered-byte integrity manifest.

Source font files and proprietary binaries are not distributed. The current uploaded UGTS reference may be retained separately for source comparison; its equations and selected mechanisms are integrated into this volume instead of concatenating unrelated older transcripts into the kernel package.

Primary implementation references

Consulted 14 September 2026. These external references support the new satnav model and hardware contract; they are separate from the user's source corpus.

- G1 ESA Navipedia, J. Sanz Subirana, J.M. Juan Zornoza and M. Hernández-Pajares, [Code Based Positioning \(SPS\)](#). Four-state model, correction terms, linearized geometry and iterative position update.
- G2 ESA Navipedia, [Satellite Coordinates Computation](#). Emission-time satellite state and reception-time frame treatment.
- G3 ESA Navipedia, [Transformations between ECEF and ENU coordinates](#). Local frame conversion and angular conventions.
- G4 ESA Navipedia, [Parameters adjustment](#). Linearized residual and least-squares framework.
- G5 ESA Navipedia, [Positioning Error](#). Formal covariance, predicted geometry and measured error distinguished.
- G6 NGA Office of Geomatics, [World Geodetic System 1984](#). Ellipsoid/reference-system definition and constants used in the coordinate utility.
- H1 NVIDIA, [GeForce RTX 50 Series Laptops](#). RTX 5070 Ti Laptop memory specification.
- H2 NVIDIA, [CUDA GPU Compute Capability](#). GeForce RTX 5070 Ti family listed at compute capability 12.0.
- H3 NVIDIA, [CUDA 12.8 Features Archive](#). Introduction of SM_120 compiler support.
- H4 NVIDIA, [CUDA Runtime API: Texture Object Management](#). Linear resource and element-read contract.

aTOMos v3.6.1.5 — UGTS satnav kernel

Tom Klootwijk NL200678942 10-07-1990

Profile: SATNAV-R1. **Release date:** 14 September 2026.

Source-grounded specialization, executable CPU and CUDA source, independent reference, finite tests and a defined UGTS observation handoff. Prepared with AI assistance.